Automated CRST Spiral Drawing Scoring Using a Vision-Transformer model: Development and Validation in a Large Essential Tremor Cohort

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Objective

The Clinical Rating Scale for Tremor (CRST) spiral drawing assessment is the most widely adopted tool for quantifying upper extremity tremor severity in essential tremor (ET). Due to coarse integer resolution (0–4), equivalent scores across patients may not reflect equivalent underlying severity. With growing volumes of focused ultrasound thalamotomy, scalable and reproducible scoring has become essential. Our objective was to score these drawings automatically, on a continuous scale.

Background

We developed a Vision Transformer model trained on a curated dataset of expert clinician-annotated CRST spiral drawings. By leveraging a transformer architecture, the model captures global relationships across the image through self-attention mechanisms. The system produces a continuous CRST rating (0.0–4.0).

Methods

Ground truth labels came from independent clinician scoring on the standard CRST 0–4 criteria. A Vision Transformer (ViT) was used because tremor severity is a property of the whole trace, not one stroke. Freezing the pre-trained backbone leaves only the classification head to learn from clinical data. The model was trained end-to-end on expert-annotated CRST drawings and outputs a continuous 0.0–4.0 score rather than integer bins. Each task—large spiral, small spiral, straight lines, handwriting—is scored separately. Evaluation covered 480 patients, 2,985 pages and 552 treatments from 2021 to 2025.

Results

Across the 552 treatments (480-patient cohort), the model produced a mean pre-treatment CRST score of 2.21 for Drawing A and 2.46 for Drawing B, with average reductions of 33% (1.3 CRST points) and 38% (1.5 CRST points) from first to last drawing.

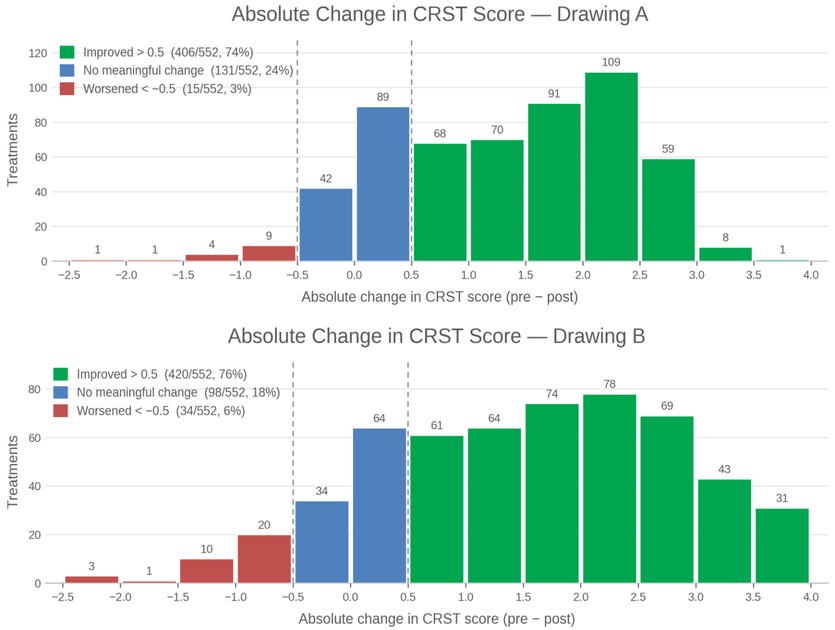


Figure 1: Absolute change in CRST score from first to last drawing — Drawing A (top), Drawing B (bottom).

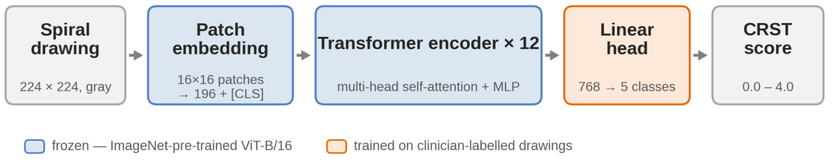


Figure 2: Vision Transformer. Patch embedding and 12 self-attention layers weigh every part of the drawing against every other; only the classification head is trained, and its class probabilities give the continuous CRST score.

For Drawing A, 74% of treatments improved by more than 0.5 CRST points and 32% by more than 2 points, whereas 3% worsened by more than 0.5 points. For Drawing B, 76% improved by more than 0.5 points and 40% by more than 2 points, whereas 6% worsened. The continuous scale resolved within-category variation that integer scoring collapses, distributing treatments within each integer band across a full range of single-decimal values.

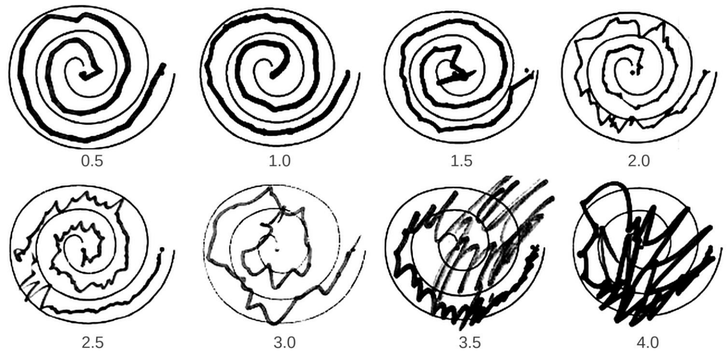


Figure 3: Example spirals at each CRST severity level, 0.5 to 4.0 in 0.5 increments, with the score beneath each drawing.

Conclusion

Validated across 552 treatments (480 patients) and 2,985 drawing pages, this AI system delivers clinician-grade automated CRST spiral drawing scoring with continuous-scale resolution, per-drawing transparency, and the scalability required for high-volume clinical programs and multi-site trials.

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